

Lab Exam:01

Title: Association/Classification/Fact and Dimension Table

Course title: Data Mining Problem Description

Course code: CSE-452
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Submitted to-

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1. Classification On my project dataset “Intelligence Automatic Supervisor Allocation System ”

The screenshot shows the Weka Explorer interface with the J48 classifier selected. The test options are set to cross-validation with 10 folds and a percentage split of 66%. The classifier output displays the J48 pruned tree structure, which is a decision tree with 4 leaves and 7 nodes. The tree structure is as follows:

```
Final_Marks <= 84
|
| Final_Marks <= 74
| |
| | Final_Marks <= 64: Poor (6.0)
| |
| | Final_Marks > 64: Average (132.0)
| |
| | Final_Marks > 74: Good (221.0)
| |
| Final_Marks > 84: Excellent (141.0)
```

The summary statistics for the stratified cross-validation are as follows:

Correctly Classified Instances	500	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0		
Root mean squared error	0		
Relative absolute error	0	%	
Root relative squared error	0	%	
Total Number of Instances	500		

The detailed accuracy by class is as follows:

TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Good
1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Poor
1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Average
1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Excellent
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	

The confusion matrix is as follows:

a	b	c	d	← classified as
221	0	0	0	a = Good
0	6	0	0	b = Poor
0	0	132	0	c = Average
0	0	0	141	d = Excellent

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The summary statistics for the stratified cross-validation are as follows:

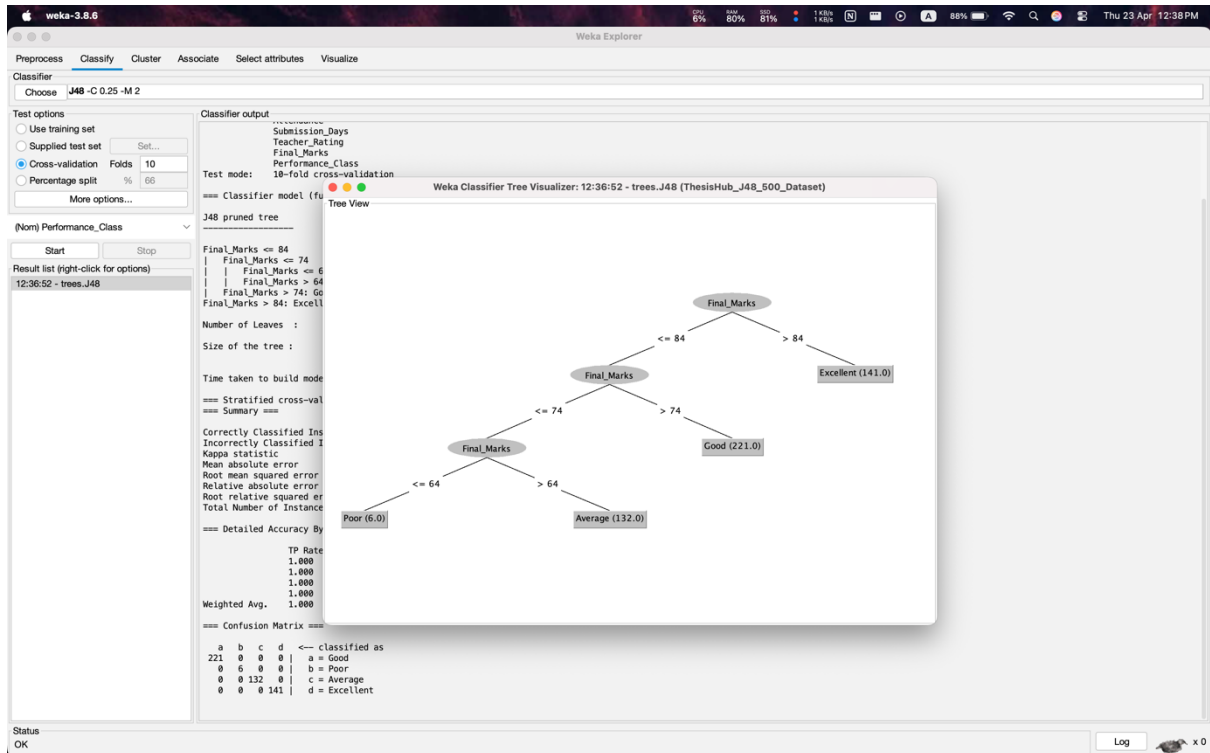
Correctly Classified Instances	500	100	%
Incorrectly Classified Instances	0	0	%
Kappa statistic	1		
Mean absolute error	0		
Root mean squared error	0		
Relative absolute error	0	%	
Root relative squared error	0	%	
Total Number of Instances	500		

The detailed accuracy by class is as follows:

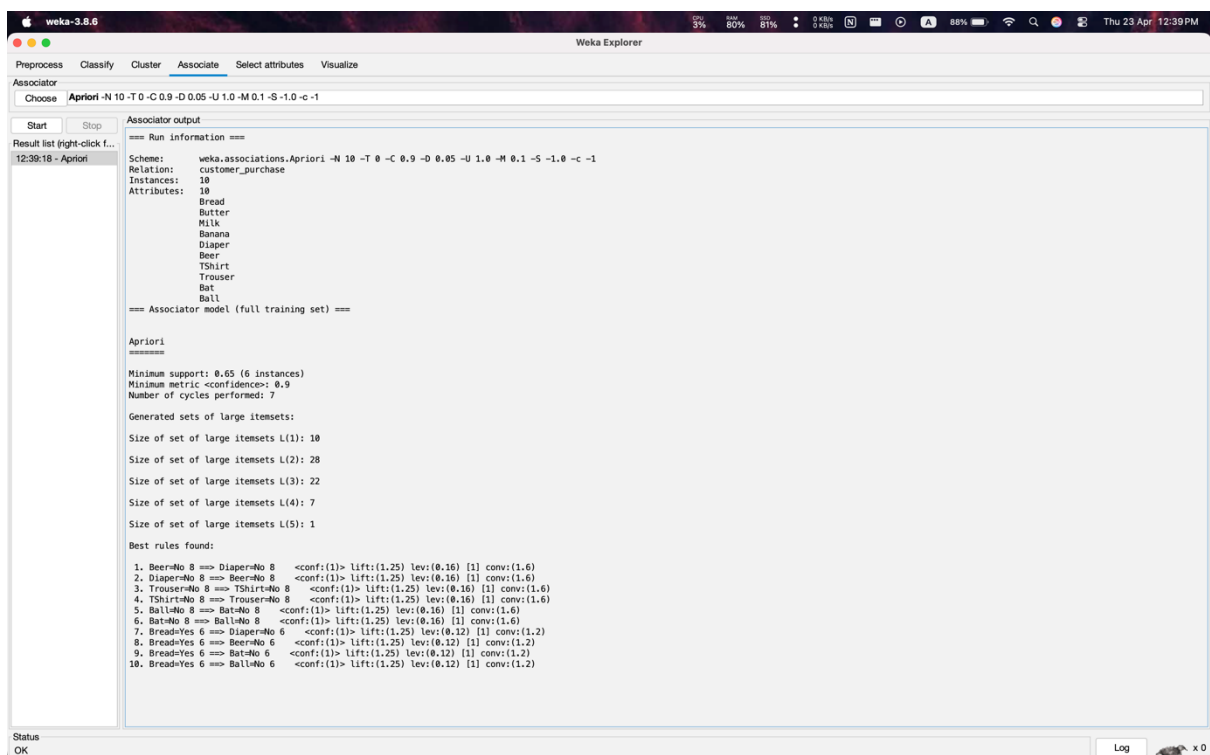
TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Good
1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Poor
1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Average
1.000	0.000	1.000	1.000	1.000	1.000	1.000	1.000	Excellent
Weighted Avg.	1.000	0.000	1.000	1.000	1.000	1.000	1.000	

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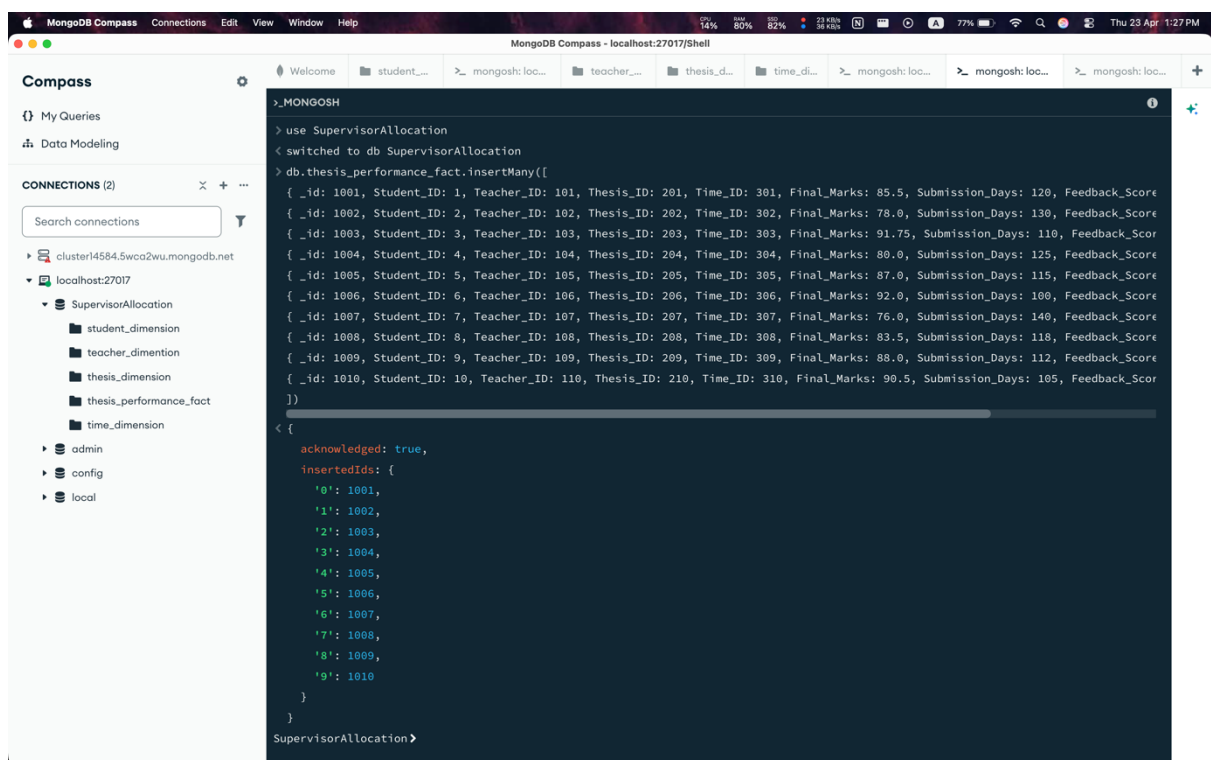


2. Association Rule On my project dataset “Intelligence Automatic Supervisor Allocation System”



3. Design Fact Table and Dimension Table

Fact Table: 1. Creation



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists two connections: 'cluster14584.5wca2wu.mongodb.net' and 'localhost:27017'. Under 'localhost:27017', the 'SupervisorAllocation' database is expanded, showing collections: 'student_dimension', 'teacher_dimension', 'thesis_dimension', 'thesis_performance_fact', and 'time_dimension'. The main panel displays the MongoDB Shell with the following commands and output:

```
> use SupervisorAllocation
switched to db SupervisorAllocation
> db.thesis_performance_fact.insertMany([
  { _id: 1001, Student_ID: 1, Teacher_ID: 101, Thesis_ID: 201, Time_ID: 301, Final_Marks: 85.5, Submission_Days: 120, Feedback_Score: 85.5 },
  { _id: 1002, Student_ID: 2, Teacher_ID: 102, Thesis_ID: 202, Time_ID: 302, Final_Marks: 78.0, Submission_Days: 130, Feedback_Score: 78.0 },
  { _id: 1003, Student_ID: 3, Teacher_ID: 103, Thesis_ID: 203, Time_ID: 303, Final_Marks: 91.75, Submission_Days: 110, Feedback_Score: 91.75 },
  { _id: 1004, Student_ID: 4, Teacher_ID: 104, Thesis_ID: 204, Time_ID: 304, Final_Marks: 80.0, Submission_Days: 125, Feedback_Score: 80.0 },
  { _id: 1005, Student_ID: 5, Teacher_ID: 105, Thesis_ID: 205, Time_ID: 305, Final_Marks: 87.0, Submission_Days: 115, Feedback_Score: 87.0 },
  { _id: 1006, Student_ID: 6, Teacher_ID: 106, Thesis_ID: 206, Time_ID: 306, Final_Marks: 92.0, Submission_Days: 100, Feedback_Score: 92.0 },
  { _id: 1007, Student_ID: 7, Teacher_ID: 107, Thesis_ID: 207, Time_ID: 307, Final_Marks: 76.0, Submission_Days: 140, Feedback_Score: 76.0 },
  { _id: 1008, Student_ID: 8, Teacher_ID: 108, Thesis_ID: 208, Time_ID: 308, Final_Marks: 83.5, Submission_Days: 118, Feedback_Score: 83.5 },
  { _id: 1009, Student_ID: 9, Teacher_ID: 109, Thesis_ID: 209, Time_ID: 309, Final_Marks: 88.0, Submission_Days: 112, Feedback_Score: 88.0 },
  { _id: 1010, Student_ID: 10, Teacher_ID: 110, Thesis_ID: 210, Time_ID: 310, Final_Marks: 90.5, Submission_Days: 105, Feedback_Score: 90.5 }
])
{
  acknowledged: true,
  insertedIds: {
    '0': 1001,
    '1': 1002,
    '2': 1003,
    '3': 1004,
    '4': 1005,
    '5': 1006,
    '6': 1007,
    '7': 1008,
    '8': 1009,
    '9': 1010
  }
}
```

The output shows that the data was successfully inserted into the 'thesis_performance_fact' collection, with 10 documents inserted and their IDs listed in the 'insertedIds' field.

2. Shows data

The image displays two screenshots of the MongoDB Compass interface, showing the 'thesis_performance_fact' collection in the 'SupervisorAllocation' database. The top screenshot shows the initial state of the collection, and the bottom screenshot shows the collection after an aggregation operation.

Top Screenshot (Initial State):

The 'thesis_performance_fact' collection contains 10 documents. The fields are: `_id` (Int32), `Student_ID` (Int32), `Teacher_ID` (Int32), `Thesis_ID` (Int32), and `Time_ID` (Int).

	<code>_id</code> Int32	<code>Student_ID</code> Int32	<code>Teacher_ID</code> Int32	<code>Thesis_ID</code> Int32	<code>Time_ID</code> Int
1	1001	1	101	201	301
2	1002	2	102	202	302
3	1003	3	103	203	303
4	1004	4	104	204	304
5	1005	5	105	205	305
6	1006	6	106	206	306
7	1007	7	107	207	307
8	1008	8	108	208	308
9	1009	9	109	209	309
10	1010	10	110	210	310

Bottom Screenshot (Aggregated State):

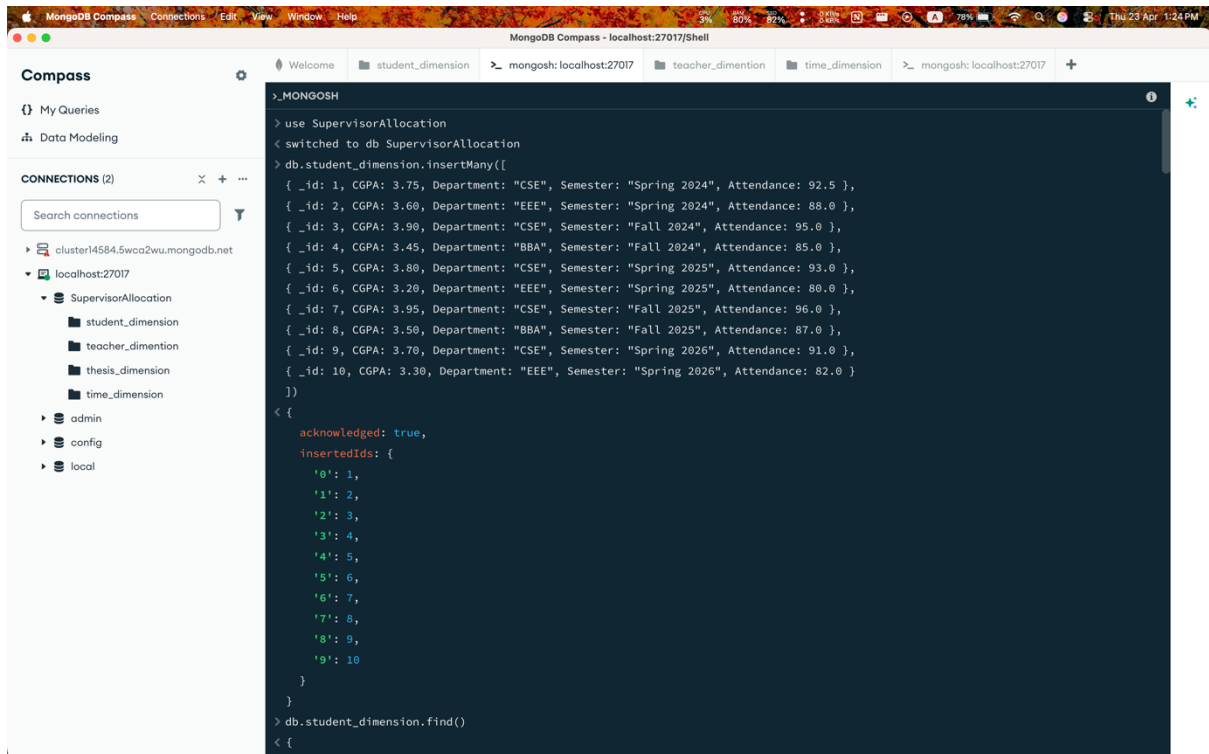
The 'thesis_performance_fact' collection contains 10 documents. The fields are: `Final_Marks` (Mixed), `Submission_Days` (Int32), `Feedback_Score` (Double), and `Task_Completion` (Int32).

	<code>Final_Marks</code> Mixed	<code>Submission_Days</code> Int32	<code>Feedback_Score</code> Double	<code>Task_Completion</code> Int32
1	85.5	120	4.6	95
2	78	130	4.2	88
3	91.75	110	4.8	98
4	80	125	4.3	90
5	87	115	4.6	93
6	92	100	4.9	97
7	76	140	4.1	85
8	83.5	118	4.4	89
9	88	112	4.2	92
10	90.5	105	4.7	96

Dimension Table

1. Creation

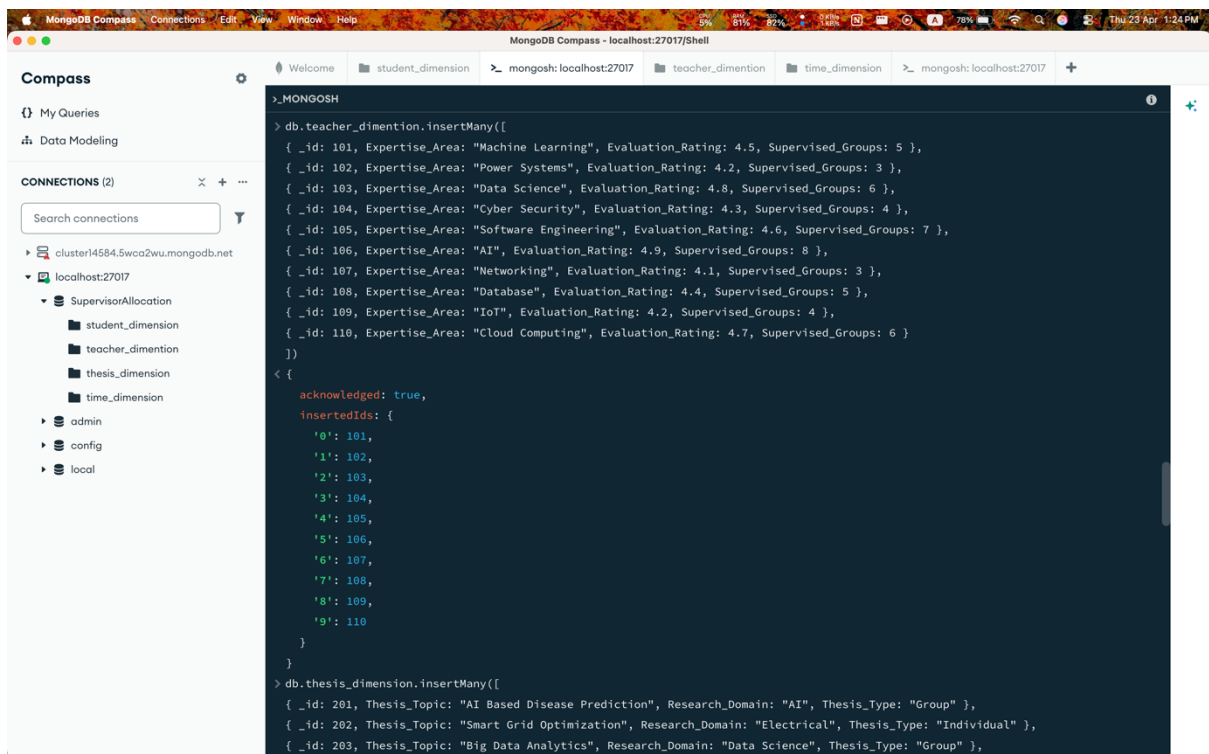
1.1 Student_dimension



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'localhost:27017' connection with a tree view showing databases like 'SupervisorAllocation', 'student_dimension', 'teacher_dimension', 'thesis_dimension', and 'time_dimension'. The main panel displays a MongoDB shell session with the following commands and output:

```
> use SupervisorAllocation
switched to db SupervisorAllocation
> db.student_dimension.insertMany([
  { _id: 1, CGPA: 3.75, Department: "CSE", Semester: "Spring 2024", Attendance: 92.5 },
  { _id: 2, CGPA: 3.60, Department: "EEE", Semester: "Spring 2024", Attendance: 88.0 },
  { _id: 3, CGPA: 3.90, Department: "CSE", Semester: "Fall 2024", Attendance: 95.0 },
  { _id: 4, CGPA: 3.45, Department: "BBA", Semester: "Fall 2024", Attendance: 85.0 },
  { _id: 5, CGPA: 3.80, Department: "CSE", Semester: "Spring 2025", Attendance: 93.0 },
  { _id: 6, CGPA: 3.20, Department: "EEE", Semester: "Spring 2025", Attendance: 80.0 },
  { _id: 7, CGPA: 3.95, Department: "CSE", Semester: "Fall 2025", Attendance: 96.0 },
  { _id: 8, CGPA: 3.50, Department: "BBA", Semester: "Fall 2025", Attendance: 87.0 },
  { _id: 9, CGPA: 3.70, Department: "CSE", Semester: "Spring 2026", Attendance: 91.0 },
  { _id: 10, CGPA: 3.30, Department: "EEE", Semester: "Spring 2026", Attendance: 82.0 }
])
{
  acknowledged: true,
  insertedIds: {
    '0': 1,
    '1': 2,
    '2': 3,
    '3': 4,
    '4': 5,
    '5': 6,
    '6': 7,
    '7': 8,
    '8': 9,
    '9': 10
  }
}
> db.student_dimension.find()
```

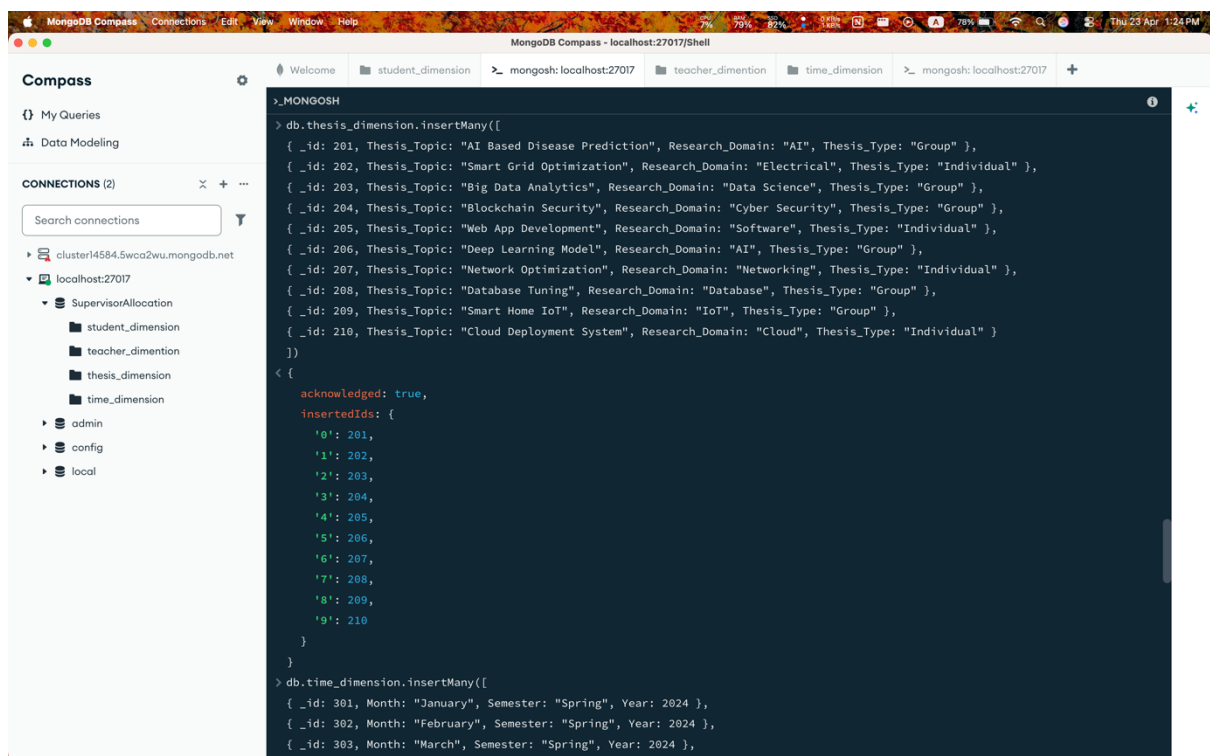
1.2 Teacher_dimension



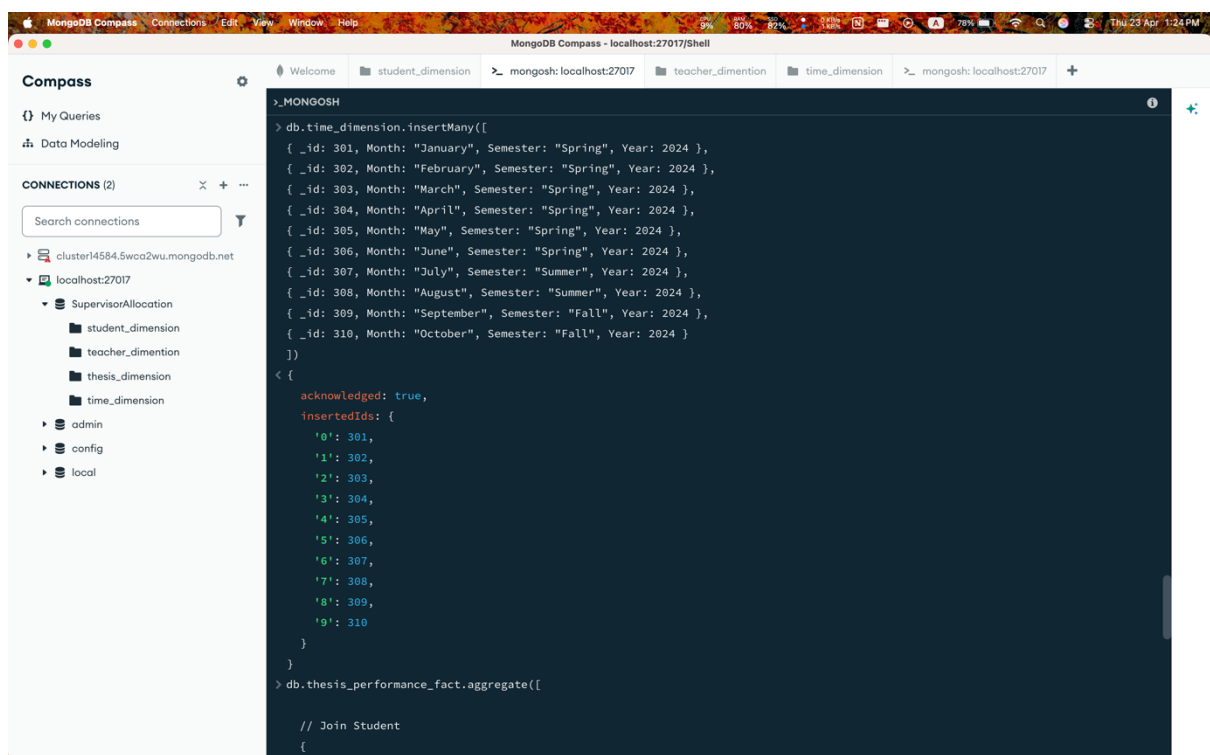
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'localhost:27017' connection with a tree view showing databases like 'SupervisorAllocation', 'student_dimension', 'teacher_dimension', 'thesis_dimension', and 'time_dimension'. The main panel displays a MongoDB shell session with the following commands and output:

```
> db.teacher_dimension.insertMany([
  { _id: 101, Expertise_Area: "Machine Learning", Evaluation_Rating: 4.5, Supervised_Groups: 5 },
  { _id: 102, Expertise_Area: "Power Systems", Evaluation_Rating: 4.2, Supervised_Groups: 3 },
  { _id: 103, Expertise_Area: "Data Science", Evaluation_Rating: 4.8, Supervised_Groups: 6 },
  { _id: 104, Expertise_Area: "Cyber Security", Evaluation_Rating: 4.3, Supervised_Groups: 4 },
  { _id: 105, Expertise_Area: "Software Engineering", Evaluation_Rating: 4.6, Supervised_Groups: 7 },
  { _id: 106, Expertise_Area: "AI", Evaluation_Rating: 4.9, Supervised_Groups: 8 },
  { _id: 107, Expertise_Area: "Networking", Evaluation_Rating: 4.1, Supervised_Groups: 3 },
  { _id: 108, Expertise_Area: "Database", Evaluation_Rating: 4.4, Supervised_Groups: 5 },
  { _id: 109, Expertise_Area: "IoT", Evaluation_Rating: 4.2, Supervised_Groups: 4 },
  { _id: 110, Expertise_Area: "Cloud Computing", Evaluation_Rating: 4.7, Supervised_Groups: 6 }
])
{
  acknowledged: true,
  insertedIds: {
    '0': 101,
    '1': 102,
    '2': 103,
    '3': 104,
    '4': 105,
    '5': 106,
    '6': 107,
    '7': 108,
    '8': 109,
    '9': 110
  }
}
> db.thesis_dimension.insertMany([
  { _id: 201, Thesis_Topic: "AI Based Disease Prediction", Research_Domain: "AI", Thesis_Type: "Group" },
  { _id: 202, Thesis_Topic: "Smart Grid Optimization", Research_Domain: "Electrical", Thesis_Type: "Individual" },
  { _id: 203, Thesis_Topic: "Big Data Analytics", Research_Domain: "Data Science", Thesis_Type: "Group" },
])
```

1.3 Thesis_dimension



1.4 Time_dimension



Show all dimension data:

1.1 Student_dimension Table

The screenshot shows the MongoDB Compass interface with the 'student_dimension' collection selected. The table contains 10 documents, each representing a student's record with fields for ID, CGPA, Department, Semester, and Attendance.

_id	CGPA	Department	Semester	Attendance
1	3.75	"CSE"	"Spring 2024"	92.5
2	3.6	"EEE"	"Spring 2024"	88
3	3.9	"CSE"	"Fall 2024"	95
4	3.45	"BBA"	"Fall 2024"	85
5	3.8	"CSE"	"Spring 2025"	93
6	3.2	"EEE"	"Spring 2025"	80
7	3.95	"CSE"	"Fall 2025"	96
8	3.5	"BBA"	"Fall 2025"	87
9	3.7	"CSE"	"Spring 2026"	91
10	3.3	"EEE"	"Spring 2026"	82

1.2 Teacher_dimension Table

The screenshot shows the MongoDB Compass interface with the 'teacher_dimension' collection selected. The table contains 10 documents, each representing a teacher's record with fields for ID, Expertise Area, Evaluation Rating, and Supervised Groups.

_id	Expertise_Area	Evaluation_Rating	Supervised_Groups
101	"Machine Learning"	4.5	5
102	"Power Systems"	4.2	3
103	"Data Science"	4.8	6
104	"Cyber Security"	4.3	4
105	"Software Engineering"	4.6	7
106	"AI"	4.9	8
107	"Networking"	4.1	3
108	"Database"	4.4	5
109	"IoT"	4.2	4
110	"Cloud Computing"	4.7	6

1.3 Thesis_dimension Table

The screenshot shows the MongoDB Compass interface with the 'thesis_dimension' collection selected. The table contains 10 documents, each with an '_id', 'Thesis_Topic', 'Research_Domain', and 'Thesis_Type'.

	_id Int32	Thesis_Topic String	Research_Domain String	Thesis_Type String
1	201	"AI Based Disease Predict.."	"AI"	"Group"
2	202	"Smart Grid Optimization"	"Electrical"	"Individual"
3	203	"Big Data Analytics"	"Data Science"	"Group"
4	204	"Blockchain Security"	"Cyber Security"	"Group"
5	205	"Web App Development"	"Software"	"Individual"
6	206	"Deep Learning Model"	"AI"	"Group"
7	207	"Network Optimization"	"Networking"	"Individual"
8	208	"Database Tuning"	"Database"	"Group"
9	209	"Smart Home IoT"	"IoT"	"Group"
10	210	"Cloud Deployment System"	"Cloud"	"Individual"

1.4 Time_dimension Table

The screenshot shows the MongoDB Compass interface with the 'time_dimension' collection selected. The table contains 10 documents, each with an '_id', 'Month', 'Semester', and 'Year'.

	_id Int32	Month String	Semester String	Year Int32
1	301	"January"	"Spring"	2024
2	302	"February"	"Spring"	2024
3	303	"March"	"Spring"	2024
4	304	"April"	"Spring"	2024
5	305	"May"	"Spring"	2024
6	306	"June"	"Spring"	2024
7	307	"July"	"Summer"	2024
8	308	"August"	"Summer"	2024
9	309	"September"	"Fall"	2024
10	310	"October"	"Fall"	2024